Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/13/2014
PI Name:	Raber, Jacob Ph.D.		
Project Title:	The Relation Between Cognitive Injury, Network Stability, and Epigenetic Change Following Exposure to Space Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	97239-3011	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	12/21/2012	**End Date:**	12/20/2016
No. of Post Docs:	2	**No. of PhD Degrees:**	4
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/20/2016 per S. Monk/HRP SR (Ed., 11/4/15)		
Key Personnel Changes/Previous PI:	Oct 2014: There were no key personnel changes in this period.		
COI Name (Institution):	Impey, Soren (Oregon Health & Science University) Turker, Mitchell (Oregon Health & Science University) Rosi, Susanna (University of California San Francisco)		
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Performance Goal Text:			

Task Description:	Space radiation poses a significant hazard to astronaut health both during the mission and at later times. However, uncertainty exists with respect to the level of risk to the CNS (central nervous system) following radiation doses relevant to those encountered during space missions. Radiation exposure in the brain affects the hippocampus, a structure critical for cognition. This proposal addresses section 1.2 of NASA Research Announcement NNJ12ZSA001N entitled, “Risk of Acute or Late Central Nervous System Effects from Space Radiation.” The proposed work focuses on changes in hippocampal network stability, including epigenetic mechanisms, in the development of cognitive dysfunction after space irradiation. Radiation exposure of the CNS significantly affects the hippocampus. Our preliminary data show impaired cognition in mice at two weeks following low dose space radiation. During spatial exploration, when an animal enters a specific location to which the cell responds (known as the cell's place field), hippocampal pyramidal neurons in the CA areas display robust activity. The place fields create an environment-specific "map," thought to be critical in establishing the context of a memory and resulting in the transcription of the immediate early gene (IEG) activity-regulated cytoskeletal associated protein (Arc). Thus, analyzing Arc provides a reliable method to detect the number of active cells forming environment-specific maps and to capture the environmental specificity of place fields. That is, place fields are highly reproducible and stable across multiple visits to the same environment. In contrast, this stability is not seen across visits in different environments. The proposed work will focus initially on effects of space radiation on the stability of hippocampal networks, assessed by analyzing the expression of Arc and other genes important for memory formation. We hypothesize that the detrimental effects of space radiation on hippocampus-dependent cognition are associated with changes in hippocampal network stability. Promoter region DNA methylation (5mC) of Arc leads to decreased expression and space radiation induces changes in DNA methylation. Moreover, DNA methylation plays a key role in regulating expression of genes required for spatial learning and memory. Thus, study of Arc expression and DNA methylation after charged particle exposure is warranted, followed by a more global search for DNA methylation changes in genes involved in learning and memory. A second form of DNA methylation, termed 5-hyrdroxymethylation (5hmC), is abundant in the brain and found at promoters of active genes, unlike 5mC, which is found at the promoters of inactive genes. In this proposal, we will test the hypotheses that the detrimental effects of space radiation on cognition are associated with Arc-related changes in the hippocampal networks involved in learning and memory and that these changes are related to space radiation-induced perturbation of 5mC and/or 5hmC. Our approach will link assessment of hippocampus-dependent cognitive changes, hippocampal network stability, and alterations in global DNA methylation and expression of genes involved in learning and memory. In Specific Aim 1, we will test the hypothesis that low-dose whole body irradiation with 56Fe or 28Si ions or protons has short- and long-term effects on cognitive performance in object recognition and spatial learning and memory requiring navigation. In Specific Aim 2, we will test the hypothesis that low dose whole body irradiation with 56Fe or 28Si ions or protons has short- and long-term effects on hippocampal networks involving Arc, and that cognitive injury following space irradiation is associated with changes in hippocampal network stability. In Specific Aim 3, we will test the hypothesis that cognitive changes following space irradiation are associated with alterations in DNA methylation and expression of Arc and other genes involved in learning and memory.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Humans are exposed to ionizing irradiation under a variety of conditions associated with medical treatment. While the risks of developing cognitive impairments in those situations are fairly well established, uncertainty exists with respect to other types of radiation exposure, such as those experienced in the space environment. There is a critical need to establish the level of risk to the CNS following radiation doses relevant to those encountered during space missions. NASA has identified four categories of radiation risk in the Integrated Research Plan (IRP) and their associated research and technology gaps. This project is pertinent to 1.2 “Risk of Acute or Late Central Nervous System Effects from Space Radiation” and to Space Radiation Risks CNS1, CNS2, CNS6, Degenerative 1, and Degenerative 2. The project focuses on the role of changes in hippocampal network stability and epigenetic mechanisms involved in these changes in the development of cognitive dysfunction after space irradiation. CNS-related topics expressly targeted by the IRP and that are specifically or indirectly addressed in the current project include early and late effects of space irradiation on cognitive function and cellular and molecular biomarkers that might be associated with cognitive injury. Innovative technologies are being developed for integrated functional assays to assess effects of space radiation on brain function.
Task Progress:	In Yr 2, we finalized the analysis of the novel object recognition data of mice irradiated at BNL and tested 2 and 20 weeks after irradiation. The mice were all 6-month-old male C57Bl6/J mice (no retired breeders). The doses of irradiation were: 56Fe (600 MeV/n; 0.1, 0.2, and 0.4 Gy); 28Si (600 MeV/n; 0.3, 0.6, and 0.9 Gy); and protons (150 MeV, 1 Gy). For each radiation condition and time point, 16 mice were cognitively tested. As last cognitive test, 8 mice of each experimental radiation condition were exposed twice to the same environment and 8 mice of each experimental radiation condition were exposed to two different environments for assessments of hippocampal network stability. In addition to these 16 mice, there were 5 cage controls. For the novel object recognition test, the mice were habituated to an open field for three subsequent days. On day 4, the mice were exposed to the open field containing two objects. On day 5, the mice were exposed to the open field with two objects but one familiar object (present on day 4) was replaced by a novel one. The time spent with each object was recorded by video and hand-scored by a researcher blinded to the treatment of the mice. Our data show impairments in novel object recognition following 56Fe, 28Si, and proton irradiation. Three days after the last water maze test day, exploration of identical or different environments was used to study the stability of hippocampal networks. Eight mice from each experimental radiation condition were placed individually into a novel environment (A) and allowed to explore for five minutes. Another 8 mice of each experimental radiation condition explored environment A for 5 minutes, and 25 minutes later they were placed in a different environment (B), a circular arena 45 cm in diameter, and allowed to explore for 5 minutes (AB Paradigm). Following behavioral testing, the mice were sacrificed by cervical dislocation and the brains quickly removed. One hemisphere was quickly frozen for catFISH and fluorescence immunostaining. Using catFISH, Arc mRNA appear as discrete nuclear foci (recent transcription ~5-10 min), and/or as diffuse mRNA in the cytoplasm (earlier transcription). Nuclear and cytoplasmic Arc can be distinguished using intronic and full-length probe respectively labeled with digoxigenin or fluorescein. Arc staining was classified as: a) None (no Arc staining); b) Arc-foci (containing only the intranuclear foci transcript); c) Arc-cyto (containing only cytoplasmic staining); or d) both Arc-foci/Arc-cyto (containing both foci and cytoplasmic staining). Using the catFISH analysis, we can identify the population of neurons activated during the first (Arc-cyto), the second (Arc-foci), and both exploration sessions (Arc-foci/Arc-cyto) and if irradiation affects these parameters.

	Furthermore, because of sensitiveness of the catFISH technique, we are able to detect possible translocation problems in the movement of Arc mRNA from the nucleus to the cytoplasm. We can determine if the neurons responding (i.e. Arc+) to an initial experience (exploration of environment A) are the same neurons that respond to a second and identical experience as expected based on sham-irradiated animals (AA Paradigm), and if the hippocampal networks activated during exploration of the first environment are statistically independent from those activated by the exploration of a different environment (AB Paradigm). Our data show that 4 weeks following 56Fe and 28Si irradiation, the percentage of Arc-positive cells is not significantly different in the following exposure to the two distinct environments as compared to exposure to twice the same environment as seen in sham-irradiated mice. Finally the omics-related changes in DNA methylation are tissue specific. We compared DNA methylation levels and metabolomics in the hippocampus and left ventricle 20 weeks following proton irradiation (150 MeV, 1Gy). As expected, 5mC was enriched in poorly expressed regions of the genome. Tissue-dependent DNA methylation, as well as DNA methylation shared in the left ventricle and hippocampus, was revealed. The gene methylation data in the left ventricle reveal significant major pathway changes that are related to known pathophysiologic processes. There was also tissue-specifc radiation-induced changes in metabolomics and in immunoreactivity of the TET2 protein, a methylcytosine dioxygenase that catalyzes the conversion of methylcytosine to 5-hydroxymethylcytosine. These data are very encouraging and indicate that all proposed outcome measures are sensitive to detect effects of relative low dose irradiation at the 2-week time point and some proposed outcome measures are sensitive to detect effects of relative low dose irradiation at the 20-week time point. These data are also encouraging as they indicate the anatomical specificity and networks involved in these effects.
Bibliography Type:	Description: (Last Updated: 10/30/2023)
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